Ratio Reasoning: Convert Measurements Between Systems

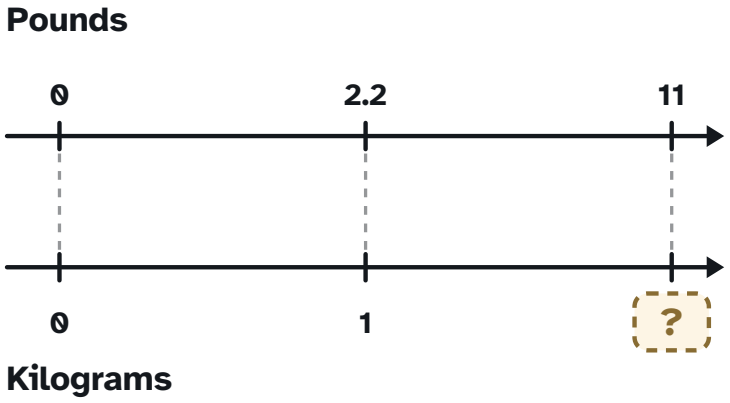
Solve a two-step problem: convert to the unit the rate uses, then apply the rate.

Name _____ **Date** _____

Tonight's spark

A pet medicine label says 6 milligrams for every 1 kilogram. A cat weighs 11 pounds. Math fact: 1 kilogram is about 2.2 pounds. About how many milligrams is the dose?

Make a guess first: is the dose more than 66 mg, or less?



Our class model: convert first, then use the rate

Try it

Work it out on paper. Write one answer on each line.

1. **A dose is 3 mg for every 1 kg. A child weighs 20 kg. How many mg is the dose?**

   A. 23 **B.** 17 **C.** 6.7 **D.** 60

   My answer _____

2. **Use 1 kg ≈ 2.2 lb. About how many kilograms is 44 pounds?**

   A. 20 **B.** 96.8 **C.** 41.8 **D.** 2

   My answer _____

3 **A dog weighs 55 pounds. The dose is 5 mg per kilogram, and 1 kg ≈ 2.2 lb. What do you do first?**

A. Multiply 55 by 5 **B.** Change 55 pounds to kilograms **C.** Divide 5 by 2.2
D. Add 55 and 5

My answer _____

4 **A dog weighs 33 pounds. Use 1 kg ≈ 2.2 lb. How many kilograms is that?**

A. 72.6 **B.** 30.8 **C.** 15 **D.** 35.2

My answer _____

5 **A child weighs 25 kg. The dose is 2 mg per kg. How many mg is the dose?**

A. 50 **B.** 27 **C.** 23 **D.** 12.5

My answer _____

Talk about it

Why must you change the unit before you use the rate?

Home mission

Find a bag of rice or flour labeled in pounds. Pretend a made-up rate: 2 mg for every 1 kilogram. Ask your student which step comes first. Real doses always come from a vet or doctor.

If we get stuck, ask

- What unit does the rate use? Are we in that unit yet?
- Which step has to come first, and why?
- What unit does your final answer have?

Do it on a screen
Scan or type the link. Same lesson, with hints.
eduwonderlab.com/lessons/3-7-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it